Fiscal Year:	FY 2016	**Task Last Updated:**	FY 03/18/2016
PI Name:	Fernandez-Pello, Carlos Ph.D.		
Project Title:	Material Ignition and Suppression Test (MIST) in Space Exploration Atmospheres		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94720-1740	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
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No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 10/31/2020 per PI and S. Olson/GRC (Ed., 3/23/16) NOTE: Extended to 10/31/2015 per NSSC information (Ed., 3/24/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX13AL10A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of NNX10AE01G, Material Ignition and Suppression in Space Exploration Atmospheres. Central Objectives: The next generation of NASA spacecraft will use lower pressures and increased oxygen concentrations than current designs. Although there is insufficient knowledge about fire behavior of materials in these Space Exploration Atmospheres (SEA), recent research suggests that material ignitability is increased in SEA. These results imply that next generation spacecraft present a greater fire safety hazard than current designs. Although the most effective fire safety strategy is to prevent ignition, if ignition does occur, it is critical to rapidly extinguish the fire. Therefore, an experimental investigation into ease of ignition and fire suppression characteristics of materials under SEA is proposed. Methods/techniques: The proposed experimental apparatus is simple and easily incorporated in the Combustion Integrated Rack on the International Space Station (ISS). It consists of a small-scale cylindrical flow duct that has fuel samples attached to its inside wall. A cylindrical radiant heater is located at the center of the duct. An oxidizer gas flows at down the length of the tube. Electrical igniters are placed downstream of each fuel sample. Time to ignition and the time between introduction of suppression agent and extinction are measured as a function of environmental conditions. The normal gravity and microgravity experimental programs will be supported by a comprehensive model of piloted ignition and extinction of materials in SEA. Microgravity experiments will be used to validate the model, which in turn will be used to predict the fire behavior of different combustible materials in SEA. Perceived Significance: Currently, there exists no testing methodology specifically designed to determine the fire hazards of materials under those conditions. The proposed work will fill that void and provide additional information about the effect of SEA on the flammability and suppression of materials. It will also provide guidance to interpret and extend current NASA testing procedures to SEA.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The project is intended to enhance the ability to predict ignition, suppression, and spread characteristics of materials in their end use configuration. This approach is called performance based design in terrestrial fire protection engineering. Computational support for this approach requires detailed information about material thermal and combustion properties. Our work on MIST will use experimental data to validate computer models attempting to predict fire behavior using material property estimation techniques. These techniques are currently a focus of research in fire protection engineering.
Task Progress:	As a part of the MIST campaign, the effects of external radiant heat flux and material opacity on time to ignition, total mass loss, and mass flux at ignition were investigated via experiments conducted with clear and black 1.27 cm diameter, 7.62 cm length cast PMMA (Poly(methyl-methacrylate) rods. It was found that clear PMMA rods exhibit longer times to ignition than do black PMMA rods for similar experimental conditions. It is hypothesized that this effect is caused by the inclusion of carbon black in the black PMMA rods. This changes the absorbency spectrum of the PMMA in the near-infrared (0.7 to 5 μm), leading to the thermal irradiation being absorbed much more closely to the surface of the samples. This in turn means that for the black PMMA rods, the material near the surface heats up more quickly than in the clear PMMA where thermal radiation absorption also occurs at depth. Thus, in the black PMMA samples, pyrolysis occurs preferentially near the surface of the sample, meaning that for a given heat flux, there is less transport time associated with achieving a flammable mixture in the gas-phase. In the case of mass flux at ignition, there is no discernible trend in the data as a function of external radiant heat flux. Nor should a trend in mass flux be expected as the gas-phase chemistry at the pilot is independent of the solid heating as long as a flammable mixture is present. Thus across the range of radiant heat fluxes tested, an observed mass flux of between 0.5 and 0.9 g m-2 s-1 corresponds very well to the theoretical range from De Ris's seminal model of 0.7614 and 0.9021 g m-2 s-1. Concurrently, the opposed flow flame spread experiments conducted aboard the International Space Station (ISS) with clear and black PMMA cylindrical rods as a part of the BASS-II campaign have helped elucidate the effects that oxygen concentration and opposed flow velocity have on flame spread over rod-like samples of thermoplastics in a micro-gravity environment. In particular, it is seen that increasing oxygen concentration or opposed flow velocity acts to increase the flame-spread rate for all three rod diameters tested. Yet, decreasing rod diameter increases flame-spread rate, an effect that can be at least in part be attributed to the curvature of the sample surface, and resultant effects of geometry on both the gas-phase boundary layer and on the solid phase conduction. To extend the results of the BASS-II experiments, and in preparation for the SoPHIE campaign of ISS bound combustion experiment, the effects of oxygen concentration, external radiant heating, and sample diameter on flame-spread rate over cast black and clear PMMA rods has been also investigated. It was found that flame spread rate decreased with decreasing oxygen concentration or eternal radiant heat flux, but increased with decreasing sample diameter. It was also found that with the use of external radiant heating, the effective Limiting Oxygen Index (LOI), or oxygen concentration at which sustained flame-spread was possible, could be reduced. Yet, only the thickest samples were able to sustain flames near or below the LOI of PMMA with the aid of the external radiant heating. It is hypothesized that this is a joint effect caused by the thermal mass of the thicker samples and the fact that the smaller, less radiant flames at lower oxygen concentrations can stand in the downstream shadow of the thicker samples, where for the thinner samples, this does not seem to be an option for the flame. In comparing the BASS-II micro-gravity results to those obtained in 1g, it is clear that flame spread in micro-gravity is faster if one accounts for the fact that the flow velocities tested in both cases are near the lower bound of what is feasible or relevant. Similar trends in flame-spread rate with sample diameter, oxygen concentration, and flow velocity (beyond the natural convection break-point) were observed, but for the tested conditions, flame-spread in μg is categorically faster than in 1g.
Bibliography Type:	Description: (Last Updated: 02/15/2026)

Abstracts for Journals and Proceedings	Link S, Fernandez-Pello C, Olson S, Ferkul P. "Flame Spread Over PMMA Cylinders in Opposed Flow with Variable Oxygen Concentration under 1g and µg Conditions." 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. 31st Annual Meeting of the American Society for Gravitational and Space Research, Alexandria, VA, November 11-14, 2015. , Nov-2015
Dissertations and Theses	Link S. (Shmuel Link) "Piloted Ignition and Flame Spread Over Clear and Black PMMA Cylinders in Opposed Flows." PhD Dissertation, University of California Berkeley, December 2015. , Dec-2015